

HR Analytics
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Lecture 01: Introduction to HR Analytics

Dear participants, in this session, we will learn the basic things about the HR analytics. What HR analytics is, what are the types of, how HR manager makes the decision, what are the things that HR manager does, what HR managers should know. In order to use the HR analytics in his or her organization. So, it is the basic and introductory session on HR analytics. So, let us start with the content. What is the content? So, first thing that we will understand that is the Introduction to HR analytics, so that this is the first thing that is what we will discuss.

Next question that HR analytics can answer, what type of questions that HR analytics can answer, what type of questions that you can frame and then you can get the questions, answers of those questions. Next one is the types of analytics, what are the types of the analytics? So here you can see there is a three types of the analytics, four types of the analytics descriptive. Right and then what are the tools of the descriptive analytics, how you can visualize predictive analytics, right and what are the questions that HR managers should answer, types of analytics, difference between MIS and HR analytics, right. So, these are the things that we will cover in this session.

So, let us start with this introduction to HR analytics. If you are into in this field, then you might have heard some people say it is the HR analytics, human resource analytics. Some people say it is the people analytics. Some people say it is the workforce analytics. So, but meaning is the same, right.

Things that are done under these analytics is the same. So, what I suggest for all HR manager for starting purpose what they should do about this HR analytics. So, they should remember the six functions of the HR, right. So, being a in any organization you are working each manager or each HR department has to perform minimum six functions. So, what are those functions? So, first function is the recruitment.

Second function is the selection right. Third function is the learning right. So, some in some organization you will see training right. So, learning. Fourth function is the

development

right.

You need to develop the employees. Fifth function is the performance, performance measurement or performance management system and sixth function is the compensation, you need to give a salary also, right. So, related to these functions, each manager has to make a decision. So, in this particular course, we will learn the basic matrix related to these six functions, right, how you can manage the recruitment effectively by using those matrix. Next function selection, how you can manage the selection, how you can manage the learning, how you can manage the development, how you can manage the performance management system and compensation, right.

So under these three, so these three are the synonyms of the HR analytics whether you call it HR analytics, whether you call it people analytics, workforce analytics, but meaning is the same, right. So, basically you will see the matrices related to these 6 functions. So, I always say to the all HR manager whichever department that you are working, first ask what are the basic questions that you need to answer in day to day work life. So, while working what are the questions that you need to answer. So, if you are working in the recruitment department, then people may ask you how many people you should attract to fulfil the vacant positions which are there.

How you should, if you are working in the L and D department, training and development department, then you may ask what is the level of the skill that is required in your employees, right, and how they can achieve that level of skill, right, skill gap analysis that you may do it. In performance, you may ask who is the best performing, which department is the best performing, how the salary component should be decided. So, such kind of questions that you need to answer in day to day, right. So, how you can find out the solutions or answers of those questions through the analytics, that is what we will learn in this course, right, I hope various matrices that we will discuss in this course that those matrices will help you to find out the solution. So, I hope, so if somebody is saying HR analytics, people analytics, workforce analytics, so meaning is same, there is no difference in these terms, right.

People, whether you call it people analytics, HR analytics, workforce analytics, everything is same, right. So, I already said what are the things that will come. I am not saying if you want to take some other things also that also you can include like engagement is there, leadership analytics is there, motivation that is what you can case. But most of the topics that you can cover under these six functions only. So, engagement that you can bring it under the performance and motivation.

So, if you can see the learning and development and performance in compensation, both

things can address this motivation issue of the employees. So, that is how you can bring the motivation aspect under these functions. So, if you will think any aspect related to the HR analytics, you can divide or you can bring under these 6 categories. Just think about it. Now question comes, what type of questions that you can answer? So what has happened? Descriptive questions like what has happened? So what has, if you are saying your recruitment is effective, so why, what has happened, so why that recruitment is effective? So if you know the number of applications that you receive after the job posting, because of that you are saying that recruitment is effective.

So, why it is effective? What has happened? So, the number of application has come more. So, that is why we are saying. If you are saying selection is effective that you have selected the people, those are effective people. So, you are saying on the basis of quality of hire, right, that people that you have hired that is good, they are good.

So, that is why. And why it is happening? Where it is happening, how it is happening and what is the underlying cause whatever is happening, right. So, such kinds of questions that HR analytics can answer for the HR manager. HR analytics will help you to find out the answers of such kind of questions. So, let us understand the types of analytics. So, so that you will understand which type of analytics will help you to answer which type of questions.

So, let us understand. So, four types of the analytics that we discuss in any. So, so four types of the analytics. So, first type is the descriptive analytics. second one is the diagnostic analytics, third one is the predictive analytics, and fourth one is the prescriptive analytics, right? So in this course we will focus more and more on this descriptive analytics, right.

So, all 60 sessions that you will see in this course focus on descriptive analytics, diagnostic, predictive and prescriptive. After this course, we will develop a new course in that we will focus on diagnostic, predictive and prescriptive. But in this course you will see we are focusing only on descriptive. So, I suggest the all HR manager before going to this diagnostic, predictive and prescriptive. First, implement this descriptive statistics in your department and then think about diagnostic, then think about predictive and then go for the prescriptive.

Because, the moment you will move from descriptive to diagnostic, little bit more statistical knowledge that you required. The moment you will go diagnostic to predictive, then more advanced techniques that you have to learn. And when you will go predictive to prescriptive, then you have to learn this mathematical modelling and you have to learn operational research tools and technique also. So, I can say that when you move from

descriptive to prescriptive, your complexity will increase and currently, most organizations they are not have the sufficient things which is required to develop an effective HR analytics department. So that is why I recommend in the initial stage, you start using these statistics using the analytical tools you start using, which are descriptive.

You should use the descriptive analytics more and more in the initial stage. The moment you feel you have developed the framework for this descriptive analytics, then you should, then only you should move towards the diagnostic and predictive analytics, right, because until or unless you have the knowledge of this statistical tool and techniques, you do not have this database sources of the data in your organization to collect the right kind of data which is required and you do not have capability to transform the data, right. So, if you do not have such kind of capabilities then think about developing these capabilities and after that you implement the all types of the analytics, right. So, in detail, we will discuss what type of capabilities is required to develop an effective HR analytics unit in the So, in detail, we will discuss, but as of now, you understand if you do not have an effective team, then do not think about the advanced level of HR analytics.

You start with descriptive. So, I hope, so descriptive will answer what kind of questions, what has happened and what is happening, right? And diagnostic will answer why did it happen? Whatever has happened, why it happened, so that relationship-related things or difference-related things that is what this diagnostic analysis will answer. So, what are the tools and techniques that will be used under each category, we will discuss in upcoming slides. Now, third type of analytics, predictive analytics will tell what will happen in the future, right. So, recruitment that you said this year 1000 people, 1000 applications that you have received. So, next year how many you will receive that you can predict through the predictive analytics.

This year you selected 100 employees, next year how many employees that you will select that you can predict through the predictive analytics. Prescriptive, so here just you can understand, right, if you have to hire 200 employees, right, and then how many HR manager you need in the organization. If you want to identify how many number of HR managers is required to hire 200 employees, so in this case you can use the prescriptive. So this is the optimisation problem, right, to hire the 200 employees how many HR managers is required, effective number. So through the optimization you can find out the solution for this prescriptive analytics.

So that is how you can use this different type of the analytics to answer these type of questions. So I already say if you do not have enough capability to use this analytics in detail then you can start with descriptive one. So, I hope this all HR manager understand, so in which stage you are, what kind of capabilities you have. So, accordingly, you can

decide the level of analytics that you will implement in your department, but this four type of analytics that exist in data science. Next, so let us understand what are the tools and technique that you can use for the descriptive analytics.

So, first thing that you can use average, simple average. So, in the coming sessions that you will see I will use, I will calculate various types of averages related to recruitment, selection, performance, and compensation. So simple average that you can understand what if you want to know how young your department is, right. So simple thing that you can do, you can calculate the average age of your department employees. So, if you can calculate the average age of your employees in the department, in whichever department that you are working, that will tell you how young your department is, right.

So, whether it is young between 20 to 25, 25 to 30, 30 to 35, 40, 50, what is the age that you can see? In the same way you can calculate the average salary. The average salary is what you can calculate, and then you can understand. The same way, you can; if you want to calculate the gender pay gap, then you can calculate the salary for males, and you can calculate the salary for females. So that is how you can calculate the various types of averages. That is what we will discuss in upcoming sessions related to performance, selection, compensation, and recruitment, right? Learning and development and then you can make various decisions.

So, this is the descriptive analytical tool first. Second one that you can see is standard deviation. So, deviation from the mean. So, whatever so that if you want to understand the outlier, how many outliers are there, right? So, deviation from the mean. So, what is the average mean and how much is the deviation? So, if deviation is high then you can say that data variation is there.

Same thing that you can see in the variance also. So how much variance is there in the data, how much it is varying, whether it is concentrated towards the mean or it is deviated from the mean. So that is what you will understand through the standard deviation and mean. So if the standard deviation is very high in terms of your age, Then you can say that some of the young people are there in your organization and some of them are very old. If the standard deviation is very low, then you can say that the ages of the employees who are working are very close, and people are from the same age group.

If the deviation is high, then you can say that people are there in the department are not from the same age group because the deviation is there, the standard deviation is very high. Same that you can use the mode. Mode is frequency, right? So whichever number is coming, so if you want to know from which age category maximum number of people are there, right? So mode that you can calculate certain age categories that you can say. So let

us assume 25 to 30, 30 to 35. So, from which age category maximum number of people are there.

So, now you can see the mode that is what you can see from which age category, which age group most of the people are there. Simple counts that you can count like number, number of applications, number of selected candidates, and number of rejected candidates. So, a simple count that you can have can also give you the number, and based on these numbers, you can calculate the ratio, you can calculate the percentage, you can calculate percentage ratios, right and then you can present effectively. So, these are the descriptive analytical tool that you can use in your HR related functions to make the decision. So, if you are using this kind of tools and techniques then you can say you are using descriptive analytics in So, I hope these are the simple tools and techniques that each HR manager would be able to interpret.

So, that is why I was saying you should focus first on descriptive analytics and then you think about the advanced one. So, these are the calculations that you can do related to the performance, compensation, learning, development, recruitment and selection. These all concepts that is what you can use and then you can do the calculation in these functions and then you can make a decision related to it, right. And now let us come to the visualization from the descriptive analytics perspective. So here you can make the histogram, so histogram that you understand, that you can make the graphs and in detail, you will understand in upcoming sessions where I will discuss various types of the histogram related to recruitment, selection, performance, compensation, learning, and development.

Right, by using all three visualisation tool. You will see I have used Tableau, I have used Power BI, I have used this Excel. By using these all three tools you can make histograms. It is not necessary you have to use only Power BI and Tableau or Excel, any tool that you can use in order to visualize this data, whichever you are comfortable with. But in this course, we will be learning all three tools to visualize the data. So, when you are using descriptive analytics, you can use this histogram to visualise the data; one of the tools is there.

The next thing you can say is to show the proportion of the things. So let us take the recruitment. You want to show the sources of recruitment. So internal versus external. So what is the percentage of the internal sources through which you have received the application and what is the proportion of the external through which you have received the number of application, right.

So, that you can put it through the pie chart, right. Next thing you can use the bar graph.

The bar graph is what you can make, right? So, bar graph that is what you can make to show number of employee, how many employees are there in various department, what is their gender, what is their salary, what is their performance level, performance rating. So, these are the things that you can show through the bar graph. So, you have understood in descriptive analytics, which analytical tool that, which statistical concept that you can use and how you can visualize that particular data.

Now, let us move to the diagnostic. So, in diagnostic first let us understand what kind of questions that you can answer. So, initially I can suggest you can start with descriptive and then slowly slowly you can move towards the diagnostic also. So, I hope HR manager has this much knowledge to implement this diagnostic after the descriptive, but start with first descriptive and then go for the diagnostic. So, what type of questions that you can answer? Why did we are observing occur? Whatever is happening, so why we are observing this? Where did it occur? So, where it is happening? Is the matrix we are monitoring related in any way to the things that we have collected the data for. So whatever data that we have collected related to the dependent variable.

So, is there any way it is related to the independent variable? Because if independent variable and dependent variable are not related then what kind of relationship that you are calculating? If you are calculating the relationship between number of application right that you received and number of people who left the organization. So, are you able to establish the relationship through any logic? Is there any logic? So, if that logic is missing between these two independent and dependent variable in which you are trying to explore the relationship that is what you need to think of, right. And if logic is there and then what is the strength of the relationship. So, you already know the correlation value, right, 0 to 1 and plus 1, plus 1 to 1, minus 1 between 0, right. So, in between somewhere you may say, you may get, so correlation value could be positive and negative also.

How much of the variability of our matrix is accounted from the data that we have collected? So, whatever data we have collected, how much variability has our matrix accounted for? So, such kind of questions that, answer of such kind of questions that you can get through the diagnostic analytics. Now, let us come to next aspect, what are the tools and technique that we used in diagnostic analytics. So, in diagnostic analytics that you can see from numbers, from descriptive numbers, means, mode, median, standard deviation, and variation from here, we have moved to some advanced level tools. So, here you can see that we will calculate correlation and regression. Analysis of variance is ANOVA, right, ANOVA is what we will, so one category ANOVA, two category ANOVA, right, and t-test also we can say, t-test, so t-test, ANOVA, correlation, regression, factor analysis, cross-tabulation, principal component analysis, correspondent analysis, multiple correspondence analysis.

So, such kind of analysis that we use under the diagnostic analytics. In this course, we will learn only about descriptive analytic tool. The next course that we will develop, in that course, we will talk about diagnostic analytics in detail, right? There we will discuss about correlation, regression, analysis of variance, ANOVA, factor analysis, cross-tabulation, principal component analysis, right. So these all things that we will discuss in the next course. But as of now you understand in diagnostic analytics these are the tools that comes.

Now let us talk about the next aspect that is the visualization. So, how you will visualize these data which is related to the diagnostic analytics. So, you can use the scattered plot to visualize the data, regression plot, plot of residual, box plot, multiple density curves. So, these are the graphs that you can make, right, my, again my advice to all HR managers is before making these all graphs, please understand your variables carefully, establish some logic between those variables and then use this data visualisation tool, Excel, Power BI or Tableau to make these graphs, right. So, for diagnostic analytics you can use this scatter plot, regression plot, plot of residual, box plot and multiple density curves, right.

So, that is what you can use. Now, third type of analytics that comes that is the predictive analytics, right. So, in the case of predictive analytics what are the things? So, we are not going to cover this type of analytics in our course, right? So, next course that we will build in we will focus on diagnostic and predictive. In this course we have covered in detail descriptive analytics. So, all these descriptive numbers, mean, mode, that is what you will see in all decisions that we are making in this course related to the HR functions that we have made only on the basis of the descriptive. So, six functions that we have covered in our in this course those functions are recruitment, selection, learning, development, performance and compensation.

So, for predictive analytics what are the tools? So tools, first tool is the regression analysis. So various types of the regression that we have linear regression, curvilinear, logistic. So these are the various types of the regression. When we will develop the next course in that we will discuss in detail. Decision trees and its variance, random forest and discriminant analysis.

So these are some of the tools for the predictive analytics. So, that we will cover in the next course that we will develop. These are the visualization tools that you can use: line charts, scattergrams and correlation plots in order to visualize the predictive data. And this is the prescriptive so, if you will see very few organizations have used this type of analytics till date, right. So because for applying this analytics you need to have high

level of statistical knowledge and operational research knowledge and mathematical modeling. If you know these things then only you will be able to use the prescriptive analytics in your organization right so after the second of course then we can think about this prescriptive also right what is the difference between MIS and analytics right so if you will see this HRA, MIS in one line i can say that HR, Management Information System is the source for the data, right.

So, on a if you are having this HRIMS then related to employee you will get the all information, right. So, you may get a information which you want related to the employee age, gender, department-wise, right. But when you have to make some decisions then you have to capture the data through some matrix, right. So, you will develop some matrix and HRIMS will give you the raw data. From that raw data you will process, you will transform, you will make it meaningful by using HR matrix and then you will make a decision.

So, I can say that HRIMS is the source of raw data, right, that you are needed to make a decision. So that raw data that you can collect from the HRIMS and then you can process it. According to the HR matrix which matrix that you have to make it and by calculating these matrices you can take a certain decision related to the HR processes. So, in this case, you can say that HR analytics offers more than an HR matrix through its potential to connect HR processes and decisions with organisational performance. So, HR matrix will give you the data and then as a HR manager you will link with HR process and organization performance and then you will take a decision related to the HR function.

So, that is the HR analytics. HRIMS will give you just data and the data you need to process and then you have to make a decision. So, this is the difference between HRIMS and HR analytics. So basic questions that you might be asking to yourself, how does this HR analytics works? So one thing that I want to tell you that you should remember this LAMP model. So whatever analytical tool that you are using, right? So first thing that you should remember this LAMP model.

So LAMP model, L stands for logic. So whatever thing that you will do in HR analytics without logic, you will not do anything, right? M stands for measure. So, first, you will apply logic, and then you will measure, and then you will process it, and then you will make a decision related to that particular thing, right? So without logic, so if you are applying analytical tool without logic, that may not give you the right answer which you are looking for. So first you need to apply the logic. So if two relationships are there, how this recruitment is related with the selection, right? Effectiveness of the recruitment is related with the selection effectiveness.

If you find any logic, then apply that. Developing the measure, then developing measures means developing the HR matrix. Develop the HR metrics, process that data and use that processes, process the data or outcome of that metrics and make a decision. So, this is the one model that you can see this how HR analytics works. Second thing that you can see that like a balance scorecard, HR scorecard is also there. Whatever HR initiative that you have taken like related to the learning and development, you have initiated two new training program.

So, how it has impacted the organization processes, right, related to, for example, customer handling. So, this new training program, how it has reduced the number of customer complaints, how it has improved the process. So, you can see some process related outcome like number of customer complaints and when number of customer complaint have reduced then how it has impacted the profit of that organization. So, that is how you can develop the HR scorecard. In order to develop the logic and link the people, strategy and performance of the various departments.

So, if you have taken any initiative related to the HR department, so how it is impacting the business processes and how these processes are impacting the business outcomes. So, that through that link you can develop this HR scorecard and then you can, so you should know how this HR analytics works. So, these are the two approaches that you can think how HR analytics works. So, first and foremost important thing is, before applying any analytical tool, you should be very much clear with your logic, why you are doing this particular analysis. If you have this answer, then go ahead with this and then measure it, process it and make a decision related to that particular problem.

So, if you remember, I had suggested The way to apply in the organization, being an HR manager; whatever questions you are having, right, make a list of all those questions. After making a list of questions, make a list of variables right about which you need to collect the data, list of variables and then think about the data analysis after collecting the data. So, in these four steps, you can use this HR analytics. Next, so what does HR analytics process? So, you can see the processes related to these statistical analysis that the whatever we have discussed descriptive, prescriptive.

So, these are the processes of the HR analytics. What is required to successfully implement the HR analytics? So, first thing that I would say. The analytical skills of a person are the skills of the HR professional. So, per person who is working in the HR department they should have the analytical skills. And the second thing that information technology, so HRIMS (Human Resource Information Management System) I was talking about, so you should have this HRIMS so that you will be able to have data you will be able to capture the data and you will bring that data together and then you will be

able to analyse it.

And third thing that I would say, basic knowledge of data analytical tool and technique. So, if you are having these three things, then you can successfully you can implement the HR analytics in your department. What are the outcome of the HR analytics? So, you can see very, you will be able to understand the relationship between various processes of HR and business outcomes. So, in this case that you can see employee engagement, how it is related with the store performance. So, HR matrix are the key majors of the HR outcomes. So, what are the HR outcomes, what are the outcomes of the recruitment, potential candidates, and how many candidates have applied? So, that will be measured through the HR matrix.

How many people are selected after the selection process will be measured through the HR matrix. So, thank you. I hope you would have learnt the basic things related to the HR analytics. So, welcome to this course. Thank you.